

Sandbox FXCore and Experimental Noize FXCore dev pedal notes

Intro:

The original Sandbox FXCore pedal design was contributed by Alexander Pedals and modified by Experimental Noize to also be a development pedal for the FXCore DSP. The design is open source so it may be used and modified as desired.

NOTE THE DESIGN IS CONSIDERED A DEVELOPMENTAL/EXAMPLE DESIGN AND NOT FOR ENDUSER USE, AS SUCH IT HAS NOT UNDERGONE FCC/EU/OTHER COMPLIANCE TESTING. DEVELOPMENTAL PEDALS PURCHASED FROM EXPERIMENTAL NOIZE ARE NOT TO BE USED OR SOLD AS END USER DEVICES, THEY ARE FOR PROGRAM DEVELOPMENT AND EVALUATION PURPOSES ONLY. IF THE EXAMPLE SANDBOX DESIGN OR ANY DERIVATIVE OF IT IS USED FOR A PRODUCTION DEVICE THE PRODUCER/SELLER OF SUCH DEVICE IS RESPONSIBLE FOR OBTAINING ANY FCC/EU/OTHER REQUIRED COMPLIANCE CERTIFICATIONS.

General info:

The Sandbox FXCore pedal ("pedal") was designed to allow people to build and modify their own development and evaluation platform. The design consists of two PCBs, and upper panel or control board consisting of the potentiometers and switches used to control the FXCore and a main PCB that has active circuitry and power conditioning. The boards are connected by a 25-trace ribbon cable. The boards were designed to fit within a 1590BBM enclosure to provide some level of protection to the boards while being evaluated.

The stomp switches are both normally-open momentary type, the enable/bypass switch is connected to a D-type F/F to create the enable/bypass signal to the FXCore. This was done to both lower the number of different components used and lower the cost through bulk purchases.

Input is stereo using a TRS type plug, outputs are stereo but split across 2 jacks. The jack next to the DC input is the left output and the center one is the right output. On the bottom of the pedal are DIP switches allowing the user to set up the input for stereo or mono. If running in mono the switches feed the "tip" signal to both left and right inputs of the ADC. **Be careful with these switch as you could short the tip and ring/shield unintentionally. Always verify proper settings prior to plugging in any cable.**

There is a second DIP switch on the bottom that selects the sample rate, note that if you change the sample rate you must power cycle or reboot the FXCore as it only checks this switch at boot to set some internal dividers.

On the left side of the pedal is a USB type C connector that allows developers to connect the board to a computer and d/l programs to the FXCore. This connector is part of an RP2040-zero board programmed to appear as an HID device and operate similar to the original FXCore development board. This makes the pedal compatible with the 2026.1 release or newer of the FXCore assembler and the online assembler available at <https://fxcore.sandboxpedal.com/>

The online assembler operates in two modes, HID and file. To use HID mode the developer must be using a Chromium compliant browser (i.e. Google Chrome) and an operating system that allows HID connections from the browser. If the user's browser is not Chromium compliant or their OS does not allow it to connect to USB devices then they may operate in file mode where they can drag and drop files onto the pedal in their file browser. The pedal will appear as a USB drive for this use. See the info and help areas of <https://fxcore.sandboxpedal.com/>

Schematics:

PDF files of the two schematics are available on the github page. CSV BOMs created from the schematics are also available on github.

PCBs:

GERBER files of the PCBs are available on the github page, note that while the panel/control PCB is defined as a 4 layer board only the two outer layers, top and bottom, are used and the two inner layers are unused. The main PCB is a 4 layer design. The control board was defined as 4 layers so that a user could panelize both boards into a single submission to save on cost at a PCB house when doing small runs.

Boards were designed to JLCPCB design rules.

Flashing the RP2040-zero:

We recommend flashing the filesystem to the RP2040-zero prior to installing it in case there are any issues. You can flash the board after it is mounted to the PCB should you need to reflash it at a later date.

Hold the BOOT button on the board while plugging in the USB cable. Release the button and a drive called "RPI-RP2" should appear. Copy the file SandboxFXCore.uf2 to this drive and after it

has finished eject and unplug the board. Plug the board back in and it should now mount as "SANDBOX-FX". Flashing of the RP2040-zero is complete.

The SandboxFXCore.uf2 is a complete copy of the FLASH, this means it will completely overwrite the memory and any files on the RP2040-zero. This is useful if you edit either the boot.py or code.py file and make an error where the board will not properly boot, you can follow the above flashing procedure to recover the board to a working state BUT this also means all files are overwritten so any files such as FXCore programs, custom changes to boot.py, etc. will be lost. Make sure to always keep copies on your computer.

Declarations for FCC and EU for the Experimental Noize development pedal sold by Experimental Noize or authorized distributor:

FCC NOTICE: This kit is designed to allow:

- (1) Product developers to evaluate electronic components, circuitry, or software associated with the kit to determine whether to incorporate such items in a finished product.
- (2) Software developers to write software applications for use with the end product. This kit is not a finished product and when assembled may not be operated or be marketed as or sold for use in a finished product unless required equipment authorizations have been obtained.

Performance is subject to the following condition: This device may not cause harmful interference, and this device must accept any interference received, including interference that may cause undesired operation.

EU

The unit covered by this declaration is a development/evaluation tool which is designed to be used for research and development in a laboratory environment. The unit covered by this declaration is not intended to be a finished appliance, nor is it intended for incorporation into finished appliances that are made commercially available as single functional units to end users.

The object of the declaration described above is in conformity with the relevant Union harmonization legislation:

The unit covered by this declaration is in conformity with EU EMC Directive, 2014/30/EU (26 February 2014).

The unit covered by this declaration, when incorporating radio equipment, is in conformity with the RED Directive, 2014/53/EU (16 April 2014).